

Prompt Generation Experiment

Language and Text Generation

1. Creative Writing

Prompt

Write a short story about a student who discovers that an old notebook in his library can predict the questions that will appear in his upcoming examinations.

Generated Content

Arun was preparing for his semester examinations when he found an old notebook hidden between two books in the college library. The notebook contained questions from subjects he had not even opened yet. Surprisingly, the same questions appeared in his class assignments the next day. Curious, Arun used the notebook wisely—not to cheat, but to prepare better. He shared his knowledge with his friends and helped them succeed. Eventually, he realized that hard work mattered more than predictions and decided to rely on his own efforts. The magical notebook taught him that knowledge becomes valuable only when it is shared.

2. Summarization

Prompt

Summarize the following paragraph in three sentences: Artificial Intelligence is transforming industries such as healthcare, education, and finance by automating tasks and improving decision-making processes. It enables faster data analysis and helps organizations provide better services. However, ethical considerations such as privacy, fairness, and transparency must be addressed for responsible AI adoption.

Generated Content

Artificial Intelligence is improving various industries through automation and better decision-making. It enables faster and more efficient data analysis. Ethical issues like privacy, fairness, and transparency are important for its responsible use.

3. Translation

Prompt

Translate the following English sentence into French: "Technology has made learning more accessible for students around the world."

Generated Content

French Translation:

"La technologie a rendu l'apprentissage plus accessible aux étudiants du monde entier."

4. Dialogue Generation

Prompt

You are a helpful virtual assistant for college students. Respond to the following query: "I am feeling stressed because of my examinations. How can I manage my time effectively?"

Generated Content

Hello! Managing examination stress becomes easier when you follow a simple study plan. Divide your subjects into smaller topics and allocate specific time slots for revision. Take short breaks between study sessions, get adequate sleep, and avoid last-minute preparation. Consistent practice and proper time management will help you feel more confident during your examinations.

Question Answering

1. Open-Ended Questions

Prompt

Explain how cloud computing is changing the way businesses store and manage their data.

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Cloud computing allows businesses to store data securely over the internet instead of relying solely on physical storage devices. It provides benefits such as scalability, cost savings, remote accessibility, and improved collaboration among employees. Cloud services also offer automated backups and enhanced security measures, making data management more efficient and reliable for organizations of all sizes.

2. Specific Questions

Prompt

What is the largest planet in our solar system?

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Jupiter is the largest planet in our solar system. It is a gas giant known for its Great Red Spot, which is a massive storm that has existed for hundreds of years.

3. Multiple Choice Questions

Prompt

Which programming language is primarily used for data analysis?

- A) Java
- B) Python
- C) HTML
- D) CSS

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The correct answer is:

B) Python

Python is widely used for data analysis because of its powerful libraries such as NumPy, Pandas, and Matplotlib.

4. Hypothetical Questions

Prompt

What would happen if humans could communicate with animals?

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If humans could communicate with animals, our understanding of nature would greatly improve. We might learn about animal behavior, environmental changes, and ways to better protect wildlife. It could also lead to stronger relationships between humans and animals and encourage more ethical treatment of living creatures.

5. Opinion-Based Questions

Prompt

Do you think artificial intelligence will play an important role in future education? Why or why not?

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Artificial intelligence is likely to play a significant role in future education by providing personalized learning experiences and assisting both students and teachers. AI can help identify individual learning needs, automate routine tasks, and make educational resources

more accessible. However, it should complement human educators rather than replace them, as teachers provide guidance, creativity, and emotional support that technology alone cannot offer.

Code Generation

1. Code Completion

Prompt

Write a Python function to check whether a given number is a prime number.

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```
def is_prime(number):
    if number <= 1:
        return False

    for i in range(2, int(number ** 0.5) + 1):
        if number % i == 0:
            return False

    return True

num = 17

if is_prime(num):
    print(num, "is a Prime Number")
else:
    print(num, "is not a Prime Number")
```

Output:

17 is a Prime Number

2. Code Translation

Prompt

Translate the following Python code into Java.

```
def square(number):
    return number * number

print(square(5))
```

Generated Content

```
public class Main {
```

```
public static int square(int number) {  
    return number * number;  
}  
  
public static void main(String[] args) {  
    System.out.println(square(5));  
}  
}
```

Output:

25

3. Code Optimization

Prompt

Optimize the following Python code to improve its efficiency.

```
sum = 0  
  
for i in range(1,1001):  
    sum += i  
  
print(sum)
```

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The code can be optimized by using Python's built-in `sum()` function, which is more concise and efficient.

```
total = sum(range(1,1001))  
  
print(total)
```

Output:

500500

4. Code Debugging

Prompt

Debug the following Java code and explain why it produces an `ArrayIndexOutOfBoundsException`.

```
public class Main{  
    public static void main(String[] args){  
        int[] numbers = {10,20,30};  
        System.out.println(numbers[3]);  
    }  
}
```

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The array contains only three elements with indices:

- $0 \rightarrow 10$
- $1 \rightarrow 20$
- $2 \rightarrow 30$

Attempting to access `numbers[3]` causes an `ArrayIndexOutOfBoundsException` because index 3 does not exist.

Corrected Code:

```
public class Main{  
    public static void main(String[] args){  
        int[] numbers = {10,20,30};  
        System.out.println(numbers[2]);  
    }  
}
```

Output:

30

Image Generation

1. Photorealistic Images

Prompt

A photorealistic image of a sunset over the ocean with palm trees silhouetted against the sky.



2. Artistic Images

Prompt

An impressionist painting of a bustling city street with people walking under umbrellas in the rain.

3. Abstract Images

Prompt

An abstract image representing the concept of hope, using bright colors and flowing shapes.

4. Image Editing

Prompt

Change the background of this photo to a starry night sky and add full moon. OR Remove the person from this image and replace them with a cat.